

Computer Science and Engineering Department
Motilal Nehru National Institute of Technology Allahabad
B.Tech IV Semester
Operating System Lab (CSN14401)
Assignment 7

1. Write a program to implement Banker's Algorithm considering processes P0 to P4 as shown in table and check if the system is in a safe state? If it is in safe state, determine the safe sequence.

Processes	Allocation A B C	Max A B C	Available A B C
P0	1 1 2	4 3 3	2 1 0
P1	2 1 2	3 2 2	
P2	4 0 1	9 0 2	
P3	0 2 0	7 5 3	
P4	1 1 2	1 1 2	

2. Write a multithreaded program that implements the banker's algorithm. Create n threads that request and release resources from the bank. The banker will grant the request only if it leaves the system in a safe state. You may write this program using Pthreads. It is important that shared data be safe from concurrent access. To ensure safe access to shared data, you can use mutex locks, which are available in the Pthreads.